



8660 - Touched by Fire: Resolving the AGN Impact on PAH Emission in Galaxies

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NGC4151				
	1	NIRSPEC	NIRSpec IFU Spectroscopy	(1) NGC4151

ABSTRACT

Polycyclic aromatic hydrocarbons (PAHs) are ubiquitous dust grains in the interstellar mediums of galaxies. Their strong MIR emission bands are widely used as tracers of star formation, and play a crucial role in studies of the co-evolution of galaxies and supermassive black holes. Galaxies with an AGN exhibit spectra with anomalous PAH band ratios, hinting that smaller grains may be destroyed by the hard radiation from the nucleus, and

raising questions about how PAH emission can be used for diagnostic purposes in active galaxies. At the same time, the interstellar radiation field (ISRF) impacts these same PAH band ratios, and AGN galaxies have a different ISRF composition compared with normal star-forming galaxies. Previous JWST-based studies of PAHs in AGN galaxies have not simultaneously incorporated the ISRF and changes to the PAH grain size distribution (GSD) when interpreting the physical implications of these anomalous PAH band ratios. We propose to utilize archival JWST NIRSpec+MIRI IFU spectroscopy to analyze PAH spectra for a set of nearby AGN galaxies, within a self-consistent modeling framework that incorporates both potential changes to the PAH GSD from an AGN as well as the impacts from spatial variations in the ISRF. Additionally, we propose for one new NIRSpec IFU observation in the nucleus of NGC 4151 to complete a uniform set of data. This new observation totals just 3 hours of charged time. With this data set and novel modeling framework, we can identify and quantify the impact of an AGN on PAH emission.

OBSERVING DESCRIPTION

We require a single NIRSpec IFU observation in the nucleus of NGC 4151 to complete a uniform set of observations with the archival data for our sample of AGN galaxies, totaling just 3 hours.

Using 16 groups and the standard 4-dither pattern, we achieve the minimum desired SNR (≥ 10) for the PAH 3.3 μ m feature in the faintest nuclear regions (~ 20 MJy/sr) based on a galaxy spectrum template scaled to Spitzer/IRAC 3.6 μ m photometry.

Care is taken to avoid any saturation while observing this bright source. We employ the NRSRAPID readout to help avoid saturation, and we thoroughly test for saturation using the ETC. We obtain the nuclear near-infrared flux from existing (but incompatible with our science goals) NIRSpec spectroscopy for our target, and find that with our chosen observational parameters there are still >20 "groups before saturation" for any pixel on the detector.

The G395H/F290P grating/filter pair is chosen to provide full wavelength coverage and to be consistent with the archival data in our proposed sample.

We also include a matched LEAKCAL observation with the same observational parameters. This calibration data will be important for mitigating bad pixels in the same capacity as a background observation would, but without the overhead associated with a new pointing. Further, astrophysical foreground emission is insignificant compared with the target emission at these wavelengths. Importantly, the MSA FOV covers much of NGC 4151 for all allowed roll angles, making the LEAKCAL critical for subtracting off stray light from stuck-open shutters.

Proposal 8660 - Targets - Touched by Fire: Resolving the AGN Impact on PAH Emission in Galaxies

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	NGC4151	RA: 12 10 32.5760 (182.6357333d)	Proper Motion RA: -0.206 mas/yr	
			Dec: +39 24 21.06 (39.40585d)	Proper Motion Dec: -0.16299998151225736 mas/yr	
			Equinox: J2000	Parallax: 2.161E-4"	
				Epoch of Position: 2000	
	<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Galaxy</i> <i>Description=[Active galactic nuclei, Active galaxies, Seyfert galaxies]</i> <i>Extended=YES</i>				

Proposal 8660 - Observation 1 - Touched by Fire: Resolving the AGN Impact on PAH Emission in Galaxies

Observation	Proposal 8660, Observation 1: NIRSPEC											Tue Feb 10 23:00:20 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
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	Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Seyfert galaxies] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSRAPID	16	1	false	true	NONE	4	4	730.1	223631
	2	G395H/F290LP	NRSRAPID	16	1	true	true	NONE	4	4	730.1	223631